

Mashrur Mahmood

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SUMMARY

Persevering and analytical, with strong problem-solving skills developed through multidisciplinary engineering projects, competitive programming, and software development. Thrives in collaborative environments, values continuous learning, and is committed to delivering reliable, innovative solutions to real-world engineering challenges.

EDUCATION

Bangladesh University of Engineering and Technology Bachelor of Science in Mechanical Engineering • CGPA: 3.67/ 4.00 • Awards: University Merit Stipend (3 Terms)	Dhaka, Bangladesh Nov 2021 – Jun 2026
Saint Joseph Higher Secondary School Higher Secondary Certificate, Science Division • GPA: 5.00/ 5.00 • Awards: Perfect Attendance	Dhaka, Bangladesh 2020

EXPERIENCE

Industrial Attachment United Ashuganj Energy Ltd. (UAEL) Report (PDF) • Completed a 13-day industrial training program at a 195 MW combined-cycle gas-engine power plant. • Studied the operation of gas engines, HRSGs, steam turbines, gas RMS, substations, and auxiliary mechanical and electrical systems. • Acquired practical exposure to DCS/SCADA operations, plant safety, preventive maintenance, and utility-scale power plant operation.	March 2025 Ashuganj, Bangladesh
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PROJECTS

Autonomous Indoor UAV SLAM and Thermal Surveying Undergraduate Project and Thesis • Designed, assembled, and integrated an autonomous quadcopter for GPS-denied indoor navigation. • Key Components: Raspberry Pi, Pixhawk 6C, Optical Flow Sensor, Distance Sensor, LiDAR, and Thermal Imager • Frameworks & Libraries: PX4, ROS2, Gazebo, slam-toolbox, nav2, and MAVLink	Apr 2025 – Jun 2026 Book (PDF) Demo (Video)
Automated Buckling Load Determination of Slender Column Electro-Mechanical System Design & Practice • Designed and developed an automated <i>electro-mechanical</i> apparatus for real-time determination of the critical buckling load of slender columns. • Implemented Arduino Mega-based control, hydraulic actuation, and a real-time buckling detection algorithm for automated testing. • Technologies: Arduino, Load cell, HX711 ADC, Hydraulic cylinder, MOSFET, Solenoid Valve, and LCD.	Nov 2023 – Mar 2024 Poster Overview (Video) Guide (Video)
Compact Heat Exchanger for Two-wheeler Vehicles Thermo Fluid System Design • Designed and manufactured a compact heat exchanger featuring flat tubes and triangular louvered fins, achieving a higher heat transfer rate than conventional designs. • Implemented a Python script to determine the optimal fin and tube geometry, reducing overall weight while maintaining the desired thermal performance. • Processes: Laser Cutting, Sheet Metal Bending, Welding and Hydraulic Testing.	Sep 2024 – Mar 2025 Report (PDF)

EXTRACURRICULAR ACTIVITIES

Competitive Programming

Online and Onsite Contests and Problem Solving

- Competed in programming contests from the school level through university, including IOI selection contests, IUPCs, and the ICPC.
- Achieved the **Expert** title (Max Rating: 1739) on Codeforces, demonstrating strong algorithmic problem-solving and competitive programming skills.
- Solved 500+ problems across platforms, covering a wide range of data structures, algorithms, and difficulty levels.
- Best Result: Secured 40th place (*Honorable Mention*) at the ICPC Dhaka Regional, competing against top university teams.

Game Development

Unity | Godot

- Developed multiple projects using Unity 3D, including Android game development, gaining hands-on experience in gameplay programming and engine workflows.
- Contributed to a Unity-based BRTA driver training simulator for manual transmission vehicles, integrating Logitech steering wheel and gear shifter peripherals. | [Demo \(Video\)](#)
- Received the Best BUET Junior Team award in a competitive 96-hour game jam involving over 30 teams across online and on-site phases.

President

2019 – 2020

Josephite Math Club

- Organized the club's flagship event, Josephite Math Mania, managing planning, coordination, and execution.
- Coordinated seminars and academic events while leading member collaboration and task delegation.
- Managed the publication of the annual club magazine *Summation*, including its layout and visual design. | [\(PDF\)](#)

Class Representative

2025-2026

Bangladesh University of Engineering and Technology

ACHIEVEMENTS & AWARDS

- Secured an Honorable Mention, ranking 40th out of 311 competing teams at the **ICPC Dhaka Regional 2025**
- Winner of the **National Math Olympiad** and participant in National Math Camp 2018
- Awarded BUET Rising Team at **DL SPRINT 2.0**, Bengali Document Layout Analysis Competition, BUET CSE FEST 2023 [Presented Paper \(PDF\)](#)
- Awarded Best BUET Juinor Team at **BUET IEEE CS GAMEJAM 2023** [Demo \(Video\)](#)
- Achieved 5th Place at **AI Contest**, BUET CSE FEST 2022 [GitHub Link](#)

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Java, JavaScript, HTML/CSS

Frameworks: ROS2, React, Node.js

Developer Tools: Git, VS Code, Visual Studio, PyCharm, Jupyter, Docker, SolidWorks

Libraries: pandas, NumPy, Matplotlib

Game Engines: Unity, Godot

Creative Tools: Adobe Photoshop, Adobe Illustrator, Adobe Premiere Pro

REFERENCES

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Relationship: Thesis Supervisor

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